

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS, TNAS & TAAS]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	C	D	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C	A,C	A,C,D	A,C	A,D	B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.20	48.00	4.00	1.00	5.00	2.50
SECTION-II (ii)	Q.	7	8	9			
	A.	94	5	2			

PART-2 : CHEMISTRY

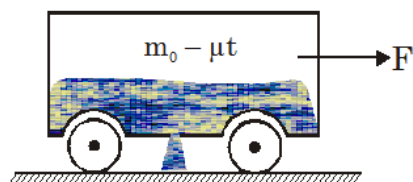
SECTION-I (i)	Q.	1	2	3	4		
	A.	D	D	B	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C,D	A,B,C	A,B,D	A,B	A	A,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	8.00	7.00	5000.00	1500.00	1.50	0.25
SECTION-II (ii)	Q.	7	8	9			
	A.	7	3	2			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	B	B	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B	B,C,D	A,D	A,B	A,B,C,D	A,B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	5.00	1.00	25.00	4.00	2.00	6.00
SECTION-II (ii)	Q.	7	8	9			
	A.	4	12	4			

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A)

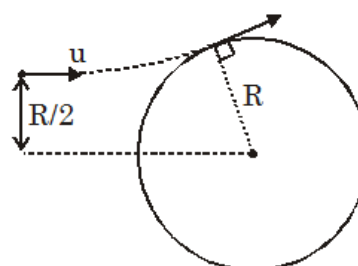


$$F = (m_0 - \mu t)a$$

$$F = (m_0 - \mu t) \frac{dv}{dt}$$

$$\int_0^v dv = \int_0^t \frac{F}{(m_0 - \mu t)} \cdot dt$$

2. Ans (C)



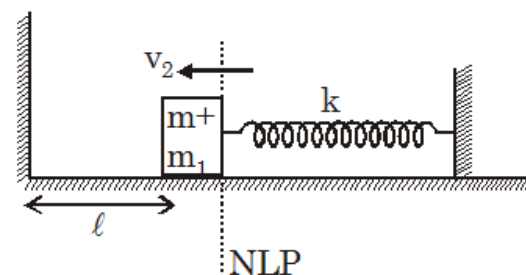
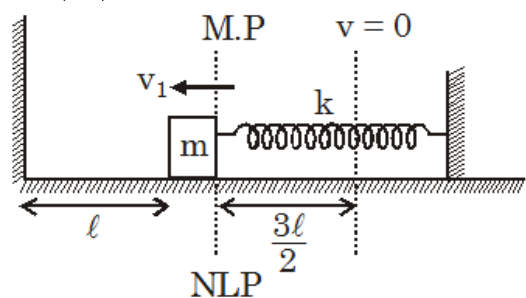
Angular momentum conservation

$$mu \frac{R}{2} = mvR \dots(i)$$

Energy conservation

$$\frac{1}{2} mu^2 + 0 = \frac{1}{2} mv^2 + \frac{kQq}{R}$$

3. Ans (D)

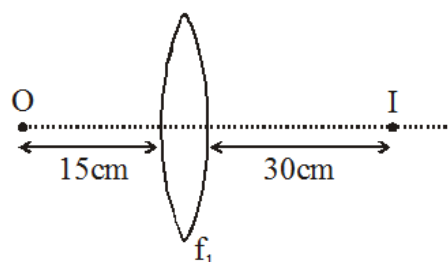


$$v_1 = \omega_1 A_1 = \sqrt{\frac{k}{m}} \cdot \frac{3l}{2}$$

$$v_2 = \omega_2 A_2 = \sqrt{\frac{k}{m+m_1}} \cdot l$$

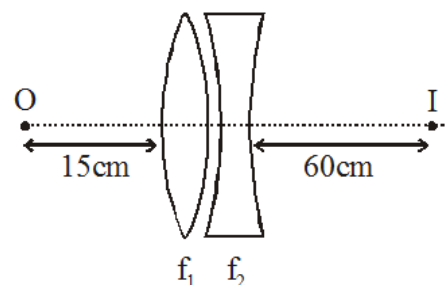
$$mv_1 = (m+m_1)v_2$$

4. Ans (C)



$$\frac{1}{30} - \frac{1}{-15} = \frac{1}{f_1}$$

$$f_1 = 10 \text{ cm}$$



$$\frac{1}{60} - \frac{1}{-15} = \frac{1}{f_{eq}}$$

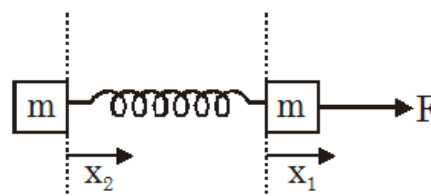
$$\text{and } \frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{12} = \frac{1}{10} + \frac{1}{f_2}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,C)



$$F = (m+m) a_{cm}$$

$$a_{cm} = \frac{F}{2m} \quad \dots(i)$$

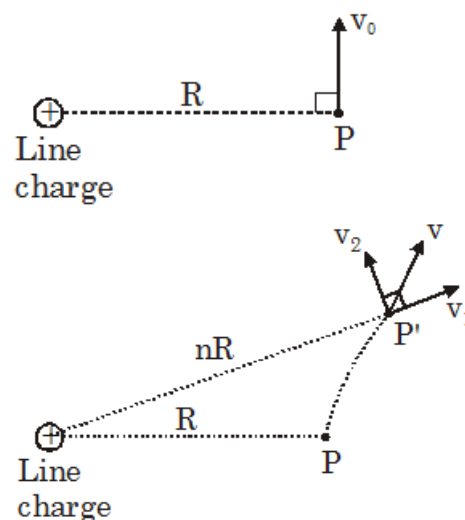
$$S_{cm} = ut + \frac{1}{2}at^2$$

$$\frac{m \cdot x_1 + mx_2}{2m} = 0 + \frac{1}{2} \cdot \frac{F}{2m} \cdot t^2$$

$$\frac{x_1 + x_2}{2} = \frac{F}{4m} t^2 \quad \dots(ii)$$

$$x_1 - x_2 = l \quad \dots(iii)$$

6. Ans (A,C)



From energy conservation

$$\frac{1}{2}mv_0^2 + Qv_p = \frac{1}{2}mv^2 + Qv_p'$$

$$\frac{1}{2}mv^2 = \frac{1}{2}mv_0^2 + Q[v_p - v_p']$$

$$= \frac{1}{2}mv_0^2 + Q \left[2k\lambda \cdot \ln \left(\frac{\eta R}{R} \right) \right]$$

$$v = \sqrt{v_0^2 + \frac{4k\lambda \cdot \ln \eta}{m}}$$

From angular momentum conservation

$$mv_0R = v_2\eta R$$

$$v_2 = \frac{v_0}{\eta}$$

$$\text{and } v_1^2 + v_2^2 = v^2 \Rightarrow v_1 = \sqrt{v^2 - v_2^2}$$

7. Ans (A,C,D)

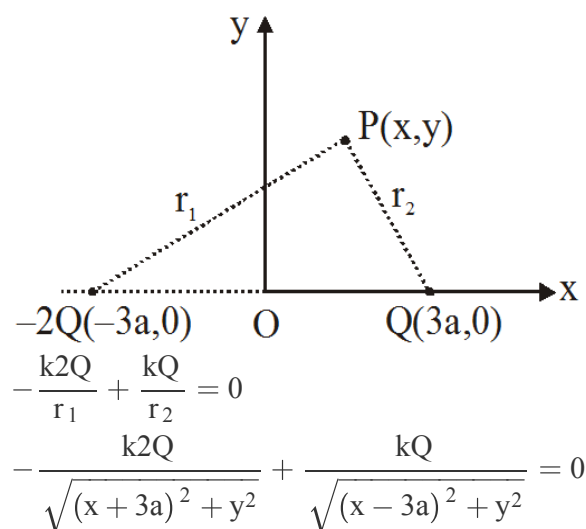
$$T = 2\pi \sqrt{\frac{I}{pE}}$$

$$= 2\pi \sqrt{\frac{\left(\frac{m\ell^2}{4}\right) \times 2}{q\ell E}}$$

In uniform $\vec{E}$ net force of dipole is always zero.

And when $\vec{p}$ is parallel to $\vec{E}$, $\vec{\tau} = \vec{p} \times \vec{E} = 0$

8. Ans (A,C)



$$x^2 + y^2 - 10ax + 9a^2 = 0$$

$$(x-5a)^2 + y^2 = (4a)^2$$

Radius of circle = $4a$

Centre of circle = $(5a, 0)$

9. Ans (A,D)

$$T_0 = 2\pi \sqrt{\frac{M}{k}}$$

$$T = 2\pi \sqrt{\frac{M+m}{k}} \quad \therefore \frac{T}{T_0} = \sqrt{\frac{M+m}{M}}$$

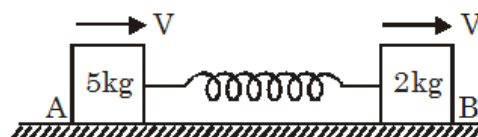
Momentum Conservation

$$Mv_1 = (M+m)V_2$$

$$M.a_0\omega_1 = (M+m)a.\omega_2$$

$$Ma_0\sqrt{\frac{k}{M}} = (M+m)a.\sqrt{\frac{k}{M+m}}$$

10. Ans (B,C,D)



From momentum conservation

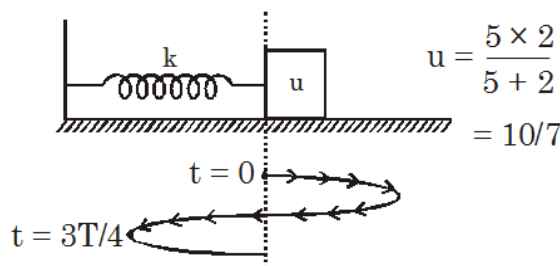
$$3 \times 5 + 2 \times 10 = 7V$$

$$V = 5 \text{ m/s}$$

From energy conservation

$$\frac{1}{2} \times 5 \times 3^2 + \frac{1}{2} \times 2 \times 10^2$$

$$= \frac{1}{2} \times 7 \times 5^2 + \frac{1}{2} \times 1120 \times x^2$$



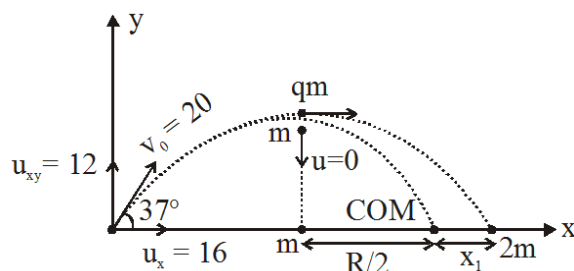
First maximum compression occurs at

$$t = \frac{3T}{4} = \frac{3}{4} \times 2\pi \sqrt{\frac{\mu}{k}}$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (1.20)



$$t = \frac{T}{2} = \frac{1}{2} \cdot \frac{2u_y}{g}$$

$$t = \frac{12}{10} = 1.2 \text{ sec}$$

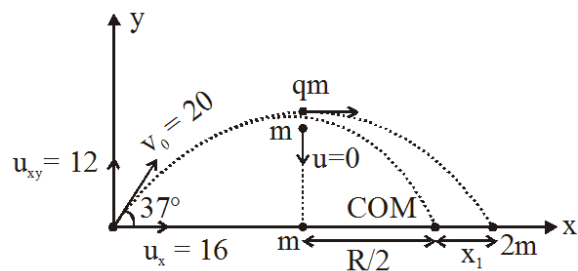
$$m \cdot \frac{R}{2} = 2m \cdot x_1$$

$$x_1 = \frac{R}{4} \quad \therefore x = R + x_1 = \frac{5R}{4}$$

$$x = \frac{5}{4} \times 2 \times \frac{u_x u_y}{g}$$

$$= \frac{5}{4} \times 2 \times \frac{16 \times 12}{10} = 48 \text{ m}$$

2. Ans (48.00)



$$t = \frac{T}{2} = \frac{1}{2} \cdot \frac{2u_y}{g}$$

$$t = \frac{12}{10} = 1.2 \text{ sec}$$

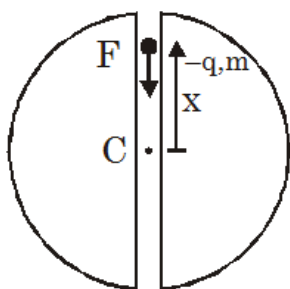
$$m \cdot \frac{R}{2} = 2m \cdot x_1$$

$$x_1 = \frac{R}{4} \quad \therefore x = R + x_1 = \frac{5R}{4}$$

$$x = \frac{5}{4} \times 2 \times \frac{u_x u_y}{g}$$

$$= \frac{5}{4} \times 2 \times \frac{16 \times 12}{10} = 48m$$

3. Ans (4.00)



$$F = qE = q \cdot \frac{kQ}{R^3} x$$

$$ma = \frac{kQq}{R^3} x$$

$$a = - \left(\frac{Qq}{4\pi\epsilon_0 m R^3} \right) x$$

$\Rightarrow$ SHM with MP at the centre of sphere

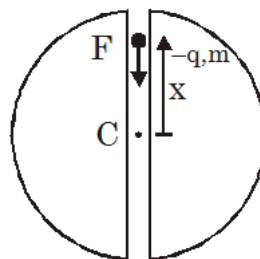
$$\omega = \sqrt{\frac{Qq}{4\pi\epsilon_0 m R^3}}$$

$$T = 2\pi \sqrt{\frac{4\pi\epsilon_0 m R^3}{Qq}}$$

$$\therefore t = \frac{T}{4} = \sqrt{\frac{\pi\epsilon_0 m R^3}{Qq}}$$

$$v = v_{\max} = A\omega = R \sqrt{\frac{Qq}{4\pi\epsilon_0 m R^3}}$$

4. Ans (1.00)



$$F = qE = q \cdot \frac{kQ}{R^3} x$$

$$ma = \frac{kQq}{R^3} x$$

$$a = - \left(\frac{Qq}{4\pi\epsilon_0 m R^3} \right) x$$

$\Rightarrow$ SHM with MP at the centre of sphere

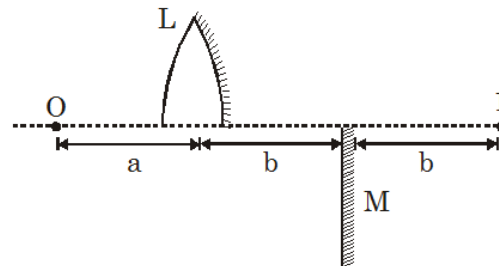
$$\omega = \sqrt{\frac{Qq}{4\pi\epsilon_0 m R^3}}$$

$$T = 2\pi \sqrt{\frac{4\pi\epsilon_0 m R^3}{Qq}}$$

$$\therefore t = \frac{T}{4} = \sqrt{\frac{\pi\epsilon_0 m R^3}{Qq}}$$

$$v = v_{\max} = A\omega = R \sqrt{\frac{Qq}{4\pi\epsilon_0 m R^3}}$$

5. Ans (5.00)



Power of silvered lens

$$P = \left(\frac{1.5 - 1}{40} \right) \times 2 + \left(\frac{-2 \times 1.5}{-40} \right)$$

$$-\frac{1}{f} = \frac{1}{40} + \frac{3}{40}$$

$$f = -10 \text{ cm}$$

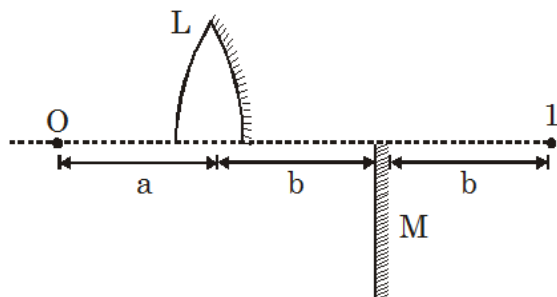
$$\text{For silvered lens } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{(a+2b)} + \frac{1}{-a} = \frac{1}{-10} \quad \dots(i)$$

$$\text{And } m = -\frac{v}{u}$$

$$2 = -\frac{a+2b}{(-a)} \quad \dots(ii)$$

6. Ans (2.50)



Power of silvered lens

$$P = \left(\frac{1.5 - 1}{40} \right) \times 2 + \left(\frac{-2 \times 1.5}{-40} \right)$$

$$-\frac{1}{f} = \frac{1}{40} + \frac{3}{40}$$

$$f = -10 \text{ cm}$$

$$\text{For silvered lens } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{(a + 2b)} + \frac{1}{-a} = \frac{1}{-10} \quad \dots(i)$$

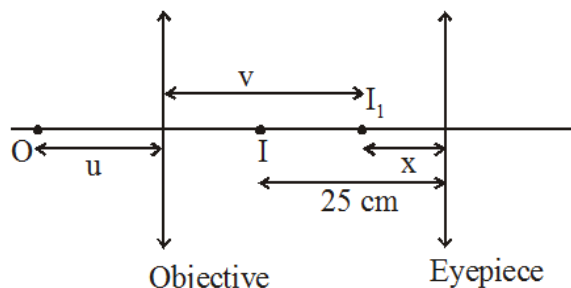
$$\text{And } m = -\frac{v}{u}$$

$$2 = -\frac{a + 2b}{(-a)} \quad \dots(ii)$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (94)



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \text{For eye piece}$$

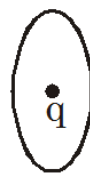
$$\frac{1}{-25} - \frac{1}{-x} = \frac{1}{5}$$

$$\text{and } v = 20 - x$$

$$\text{For objective } \Rightarrow \frac{1}{v} - \frac{1}{u} = \frac{1}{f_0}$$

$$m = \frac{v}{u} \left(1 - \frac{D}{f_e} \right)$$

9. Ans (2)



$$-m\omega^2 x = -\frac{kQqx}{(x^2 + R^2)^{3/2}}$$

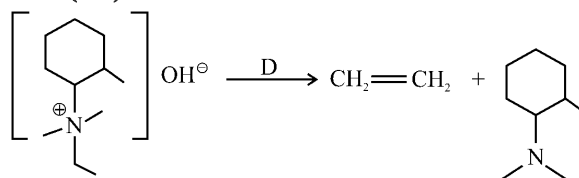
$$\omega = \sqrt{\frac{kQq}{mR^3}} = 16$$

$$\omega' = \sqrt{\frac{kQq}{m(4R)^3}} = \frac{16}{8} = 2$$

PART-2 : CHEMISTRY

SECTION-I (i)

2. Ans (D)



3. Ans (B)

Let v.p. of A = P_A^0 & v.p. of B = P_B^0

when $\eta_A = 4$; $\eta_B = 6$

$k_A = 0.4$; $\eta_B = 0.6$

so $460 = 0.4 \times P_A^0 + 0.6 \times P_B^0$

$$\Rightarrow 2P_A^0 + 3P_B^0 = 2300$$

now when $\eta_A = 16$ then

$$k_A = \frac{4}{20} \text{ \& } k_B = \frac{16}{20} = \frac{1}{5} ; = \frac{4}{5}$$

so $480 = \frac{1}{5} P_A^0 + \frac{4}{5} P_B^0$

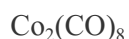
$$P_A^0 + 4P_B^0 = 2400 \dots\dots (1)$$

$$2P_A^0 + 3P_B^0 = 2300 \dots\dots (2)$$

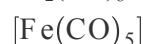
solving (1) & (2) we get $P_A^0 = 400$ &

$$P_B^0 = 500$$

4. Ans (D)



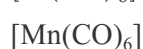
EAN of Co = 36



EAN of Fe = 36



EAN of V = 36



EAN of Mn = 37

PART-2 : CHEMISTRY

SECTION-I (ii)

9. **Ans (A)**

Because Cs^+ has large size as compare to Mg^{+2} , Ca^{+2} and Pb^{+2}

10. **Ans (A,D)**

(i) oxidation state & co-ordination number in both complexes is +3 and 6 respectively

(ii) Number of Co-N linkages in both complexes are 4

PART-2 : CHEMISTRY

SECTION-II (i)

3. **Ans (5000.00)**

reaction is zero order

$$\text{now } -\frac{1}{4} \frac{d[A]}{dt} = K_{\text{reaction}}$$

$$\text{so } -\frac{d[A]}{dt} = 4 \times K = 20 \text{ mm /sec}$$

$$\text{now at} = A_0 - K \times t$$

$$p_A = 4000 - 20 \times 100$$

$$p_A = 2000 \text{ mm of Hg}$$

$$\text{now } p_B = \frac{3}{4} \times 500 = 1500.00$$

$$\text{now } , T_{\text{final}} = P_A = 0 \text{ so}$$

$$t = 200 \text{ sec.}$$

so after

$$250 \text{ sec, } P_T = \frac{3}{4} \times 400 + \frac{2}{4} \times 4000$$

$$= 3000 + 2000 = 5000$$

4. **Ans (1500.00)**

reaction is zero order

$$\text{now } -\frac{1}{4} \frac{d[A]}{dt} = K_{\text{reaction}}$$

$$\text{so } -\frac{d[A]}{dt} = 4 \times K = 20 \text{ mm /sec}$$

$$\text{now at} = A_0 - K \times t$$

$$p_A = 4000 - 20 \times 100$$

$$p_A = 2000 \text{ mm of Hg}$$

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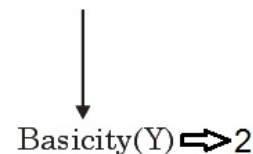
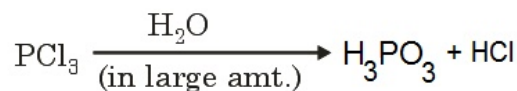
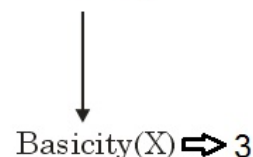
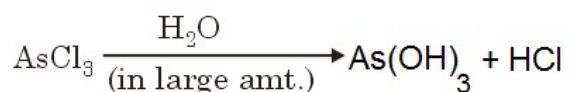
$$t = 200 \text{ sec.}$$

so after

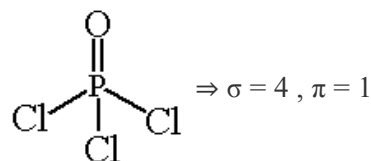
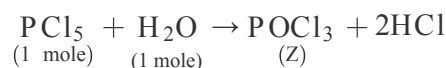
$$250 \text{ sec, } P_T = \frac{3}{4} \times 400 + \frac{2}{4} \times 4000$$

$$= 3000 + 2000 = 5000$$

5. **Ans (1.50)**



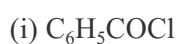
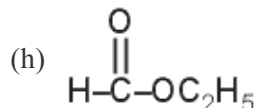
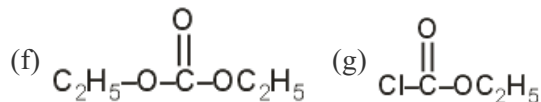
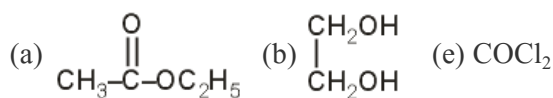
6. **Ans (0.25)**



PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (7)



9. Ans (2)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$$(cx + d)y = ax + b$$

Differentiate w.r.t. x :

$$(cx + d)y_1 + cy = a$$

$$\Rightarrow xy_1 + \frac{dy_1}{c} + y = \frac{a}{c}$$

Differentiate w.r.t. x :

$$y_1 + xy_2 + \frac{d}{c}y_2 + y_1 = 0 \Rightarrow -\frac{d}{c} = \frac{2y_1 + xy_2}{y_2}$$

Differentiate w.r.t. x :

$$\frac{y_2(2y_2 + xy_3 + y_2) - (2y_1 + xy_2)y_3}{y_2^2} = 0$$

$$\Rightarrow 3y_2^2 = 2y_1y_3 \Rightarrow \frac{y_2^2}{y_1y_3} = \frac{2}{3}$$

2. Ans (B)

Coefficient of x in f(x) is $\frac{d}{dx}(f(x))$ at x = 0.

$$\Rightarrow f'(x) = \begin{vmatrix} 2(x-1) & 3(x-2)^2 & 1 \\ [0.9em](x-1)^3 & x^2 & (x+1)^2 \\ [0.9em]x+2 & x+1 & x^5 \end{vmatrix} +$$

$$\begin{vmatrix} (x-1)^2 & (x-2)^3 & (x+1) \\ [0.9em]3(x-1)^2 & 2x & 2(x+1) \\ [0.9em]x+2 & x+1 & x^5 \end{vmatrix} +$$

$$\begin{vmatrix} (x-1)^2 & (x-2)^3 & x+1 \\ [0.9em](x-1)^3 & x^2 & (x+1)^2 \\ [0.9em]1 & 1 & 5x^4 \end{vmatrix}$$

$$= \begin{vmatrix} -2 & 12 & 1 \\ -1 & 0 & 1 \\ 2 & 1 & 0 \end{vmatrix} + \begin{vmatrix} 1 & -8 & 1 \\ 3 & 0 & 2 \\ 2 & 1 & 0 \end{vmatrix} + \begin{vmatrix} 1 & -8 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = -16$$

3. Ans (B)

$$f(|x-3|) + 2f(|x+3|) = x^2 + 2x \quad \dots(1)$$

Replace x $\rightarrow -x$

$$f(|x+3|) + 2f(|x-3|) = x^2 - 2x \quad \dots(2)$$

Solving (1) & (2)

$$\text{we get } f(|x-3|) = \frac{x^2 - 6x}{3} = \frac{(x-3)^2 - 9}{3}$$

$$\Rightarrow f(t) = \frac{t^2 - 9}{3}$$

4. Ans (C)

$$\therefore h(-1) = 4 \Rightarrow h^{-1}(4) = -1$$

$$\therefore g\left(-\frac{1}{2}\right) = -1 \Rightarrow g^{-1}(-1) = -\frac{1}{2}$$

$$\therefore f(1) = -\frac{1}{2} \Rightarrow f^{-1}\left(-\frac{1}{2}\right) = 1$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B)

$$\text{Clearly } \lim_{x \rightarrow \alpha^+} \frac{\sin([x^3] - [x]^3)}{(x - \alpha)^3} = 0$$

$\therefore$ for existence of limit L.H.L = 0.

$$\Rightarrow \alpha^3 - 1 - (\alpha - 1)^3 = 0$$

$$\Rightarrow \alpha^3 - 1 - (\alpha^3 - 3\alpha^2 + 3\alpha - 1) = 0$$

$$3\alpha^2 - 3\alpha = 0 \Rightarrow \alpha = 0, 1$$

6. Ans (B,C,D)

(A) $f(1) = \log_2(4+4) = 3 \Rightarrow g(3) = 1$

(B) $f'(x) = \frac{2^{(x^3+x)} \ln 2 (3x^2+1)}{\ln 2 (2^{x^3+x} + 4)}$

$f'(1) = \frac{4(4)}{8} = 2$

$g'(3) = \frac{1}{f'(1)} = \frac{1}{2}$

(C) $g(3) = 1$

$\Rightarrow \lim_{x \rightarrow 3} (g(x))^{\frac{\ln 4}{x-3}} = e^{\lim_{x \rightarrow 3} \left(\frac{g(x)-1}{x-3} \right) \ln 4}$
 $= e^{g'(3) \cdot \ln 4} = e^{\frac{1}{2} \ln 4} = 2$

(D) $f(x)_{\min} = \lim_{x \rightarrow -\infty} f(x) = \log_2(2^{-\infty} + 4) = 2$

so range is $(2, \infty)$

7. Ans (A,D)

$g(x) = f^{-1}(x) \Rightarrow f(g(x)) = x$

$\therefore f'(g(x)) \cdot g'(x) = 1$

$\therefore g'(x) = \frac{1}{f'(g(x))}$

Also $f(x) = -4e^{\frac{1-x}{2}} + 1 + x + \frac{x^2}{2} + \frac{x^3}{3}$

$\therefore f'(x) = +\frac{4}{2}e^{\frac{1-x}{2}} + 1 + x + x^2$

$f'(x) = 2e^{\frac{1-x}{2}} + 1 + x + x^2$

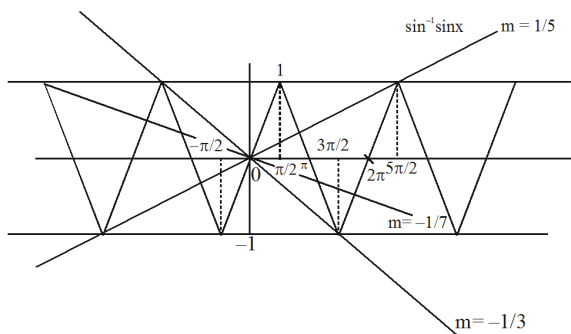
$\therefore f'(1) = 2 + 1 + 1 + 1 = 5$

and $f(1) = -\frac{7}{6}$

$\therefore g\left(-\frac{7}{6}\right) = 1$

so $g'(x) = \frac{1}{f'(g(x))}$

8. Ans (A,B)



For negative 'm' to have

five solutions $m \in \left(-\frac{1}{7}, -\frac{1}{3}\right)$

9. Ans (A,B,C,D)

$f(x) = x - 1, g(x) = \frac{1}{x}$

$f \circ g = \frac{1}{x} - 1$

$g \circ f \circ g = \frac{1}{\frac{1}{x} - 1} = \frac{x}{1-x} \Rightarrow [g \circ f \circ g] = \left[\frac{x}{1-x} \right]$

$\frac{x}{1-x}$ is discontinuous at integers $\Rightarrow \frac{x}{1-x} = n$

$\Rightarrow x = \frac{n}{n+1}, n \in \mathbb{I} - \{-1\}$

$f \circ g$ is discontinuous at $x = 1$

$\therefore [g \circ f \circ g]$ is also discontinuous at $x = 1$.

10. Ans (A,B,D)

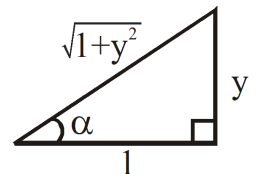
Let

$\sec^{-1}(\sqrt{1+x^2}) = \theta \Rightarrow \sec \theta = \frac{\sqrt{1+x^2}}{1}$

$\theta = \tan^{-1}x$

$\cot^{-1}\left(\frac{1}{z}\right) = \tan^{-1}(z)$

Let



$\operatorname{cosec}^{-1}\left(\frac{\sqrt{1+y^2}}{y}\right) = \alpha$

$\Rightarrow \operatorname{cosec} \alpha = \frac{\sqrt{1+y^2}}{y}$

$\alpha = \tan^{-1}(y)$

$\Rightarrow \tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$

$\Rightarrow x + y + z = xyz$

Also $\frac{x+y+z}{3} \geq (xyz)^{\frac{1}{3}}$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (5.00)**

Given

$$f(x) = \lim_{n \rightarrow \infty} \frac{x^{2n+1} \sin \frac{1}{x} + (x^2 + 2x + (\lambda + 1))}{x^{2n} + 3.x \left(\frac{1}{x} + \cos \frac{1}{x^2} \right)}$$

$$= \lim_{n \rightarrow \infty} \frac{x \cdot (x^2)^n \sin \frac{1}{x} + (x^2 + 2x + (\lambda + 1))}{(x^2)^n + 3.x \left(\frac{1}{x} + \cos \frac{1}{x^2} \right)}$$

$$f(x) = \begin{cases} \frac{x^2 + 2x + (\lambda + 1)}{3 \left(1 + x \cdot \cos \frac{1}{x^2} \right)} & \text{if } 0 \leq x^2 < 1 \\ \frac{x \sin \frac{1}{x} + (1 + 2x + (\lambda + 1))}{1 + 3(1 + x \cos 1)} & \text{if } x^2 = 1 \\ x \cdot \sin \frac{1}{x} & \text{if } x^2 > 1 \end{cases}$$

$$\therefore \lim_{x \rightarrow 0} f(x) = 2 \Rightarrow \frac{\lambda + 1}{3} = 2 \Rightarrow \lambda = 5$$

2. **Ans (1.00)**

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} x \sin \frac{1}{x} = \lim_{x \rightarrow \infty} \frac{\sin \frac{1}{x}}{\left(\frac{1}{x} \right)} = 1$$

3. **Ans (25.00)**

$$f(x^2 + y^2) = f(2xy) + f(x^2 - y^2) \quad \dots(1)$$

$$\text{Let } 2xy = a \text{ \& } x^2 - y^2 = b$$

$$f(\sqrt{a^2 + b^2}) = f(a) + f(b) \quad (\text{Let } a, b > 0)$$

$$\Rightarrow f(\sqrt{a + b}) = f(\sqrt{a}) + f(\sqrt{b})$$

$$\text{Let } f(\sqrt{x}) = g(x)$$

$$\therefore g(a + b) = g(a) + g(b)$$

$$\Rightarrow g(x) = \lambda x \Rightarrow f(\sqrt{x}) = \lambda x$$

$$\Rightarrow f(x) = \lambda x^2, x > 0$$

$$\Rightarrow f'(x) = 2\lambda x \Rightarrow 2 = 2\lambda \Rightarrow \lambda = 1$$

$$\Rightarrow f(x) = x^2; x > 0 \quad \dots(2)$$

$$\text{Put } x = y = 0 \text{ in (1)} \Rightarrow f(0) = 0$$

$$x = 0 \text{ in (1)} \Rightarrow f(y^2) = f(-y^2)$$

$$\Rightarrow f \text{ is even function}$$

$$\Rightarrow f(x) = x^2 \forall x \in \mathbb{R}$$

4. **Ans (4.00)**

$$f(x^2 + y^2) = f(2xy) + f(x^2 - y^2) \quad \dots(1)$$

$$\text{Let } 2xy = a \text{ \& } x^2 - y^2 = b$$

$$f(\sqrt{a^2 + b^2}) = f(a) + f(b) \quad (\text{Let } a, b > 0)$$

$$\Rightarrow f(\sqrt{a + b}) = f(\sqrt{a}) + f(\sqrt{b})$$

$$\text{Let } f(\sqrt{x}) = g(x)$$

$$\therefore g(a + b) = g(a) + g(b)$$

$$\Rightarrow g(x) = \lambda x \Rightarrow f(\sqrt{x}) = \lambda x$$

$$\Rightarrow f(x) = \lambda x^2, x > 0$$

$$\Rightarrow f'(x) = 2\lambda x \Rightarrow 2 = 2\lambda \Rightarrow \lambda = 1$$

$$\Rightarrow f(x) = x^2; x > 0 \quad \dots(2)$$

$$\text{Put } x = y = 0 \text{ in (1)} \Rightarrow f(0) = 0$$

$$x = 0 \text{ in (1)} \Rightarrow f(y^2) = f(-y^2)$$

$$\Rightarrow f \text{ is even function}$$

$$\Rightarrow f(x) = x^2 \forall x \in \mathbb{R}$$

5. **Ans (2.00)**

by (i) degree of $g(x)$ must be less than 4

by (iii) $g(x)$ must have $x - 1$ and $x - 2$ in its factors

by (iv) No other factor of $g(x)$ will be possible as it can not be a linear factor or a repeated factor

$$\therefore \text{Let } g(x) = a(x - 1)(x - 2)$$

$$\therefore h(x) = \frac{x^4 - 3x^3 + 2}{a(x - 1)(x - 2)}$$

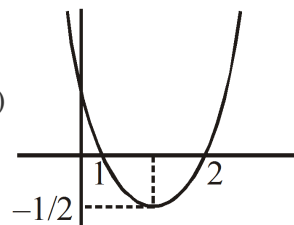
$$h(x) = \frac{(x - 1)(x^3 - 2x^2 - 2x - 2)}{a(x - 1)(x - 2)}$$

$$\therefore \lim_{x \rightarrow 1} h(x) = \lim_{x \rightarrow 1} \frac{x^3 - 2x^2 - 2x - 2}{a(x - 2)}$$

$$= \frac{5}{a} = \frac{5}{2} \Rightarrow a = 2$$

$$\therefore g(x) = 2(x - 1)(x - 2)$$

$$= 2x^2 - 6x + 4$$



$$\text{sgn}(g(x)) = \begin{cases} 1, & x < 1 \\ 0 & x = 1 \\ -1 & 1 < x < 2 \\ 0 & x = 2 \\ 1 & x > 2 \end{cases}$$

6. Ans (6.00)

by (i) degree of $g(x)$ must be less than 4

by (iii) $g(x)$ must have $x - 1$ and $x - 2$ in its factors

by (iv) No other factor of $g(x)$ will be possible as it can not be a linear factor or a repeated factor

$$\therefore \text{Let } g(x) = a(x-1)(x-2)$$

$$\therefore h(x) = \frac{x^4 - 3x^3 + 2}{a(x-1)(x-2)}$$

$$h(x) = \frac{(x-1)(x^3 - 2x^2 - 2x - 2)}{a(x-1)(x-2)}$$

$$\therefore \lim_{x \rightarrow 1} h(x) = \lim_{x \rightarrow 1} \frac{x^3 - 2x^2 - 2x - 2}{a(x-2)}$$

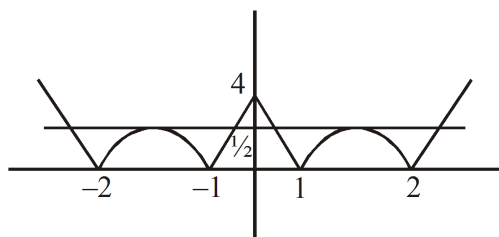
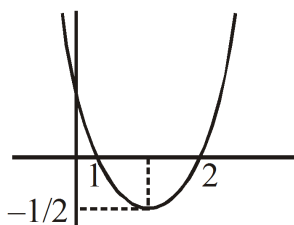
$$= \frac{5}{a} = \frac{5}{2} \Rightarrow a = 2$$

$$\therefore g(x) = 2(x-1)(x-2)$$

$$= 2x^2 - 6x + 4$$

graph of $|g(x)|$

6 solutions



PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (4)

$$([x]+[y]+[z])((x-y)^2+(y-z)^2+(z-x)^2) = 8$$

possible only when $[x] + [y] + [z] = 4$

$$[x] = 2, [y] = 1, [z] = 1$$

$$\text{or } [x] = 1, [y] = 2, [z] = 1$$

$$[x] = 1, [y] = 1, [z] = 2$$

8. Ans (12)

Let point be $P(\alpha, \sqrt{\alpha})$

$$\text{Area of } \triangle OAB = \frac{1}{2}$$

OA.OB

Slope of tangent at

$$P = \frac{1}{2\sqrt{\alpha}}$$

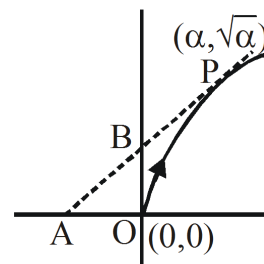
Equation of tangent

$$y - \sqrt{\alpha} = \frac{1}{2\sqrt{\alpha}}(x - \alpha)$$

$$OA = -a, OB = \frac{\sqrt{a}}{2}$$

$$\text{Area} = \frac{1}{2} | -a | \frac{\sqrt{a}}{2} = \frac{1}{2} \frac{a^{3/2}}{2} = \frac{a^{3/2}}{4}$$

$$\left(\frac{dA}{dt} \right)_{t=24} = \left(\frac{3}{8} a^{1/2} \right) \frac{da}{dt} = 12$$



9. Ans (4)

$$y_1 = \frac{dy}{dx} = 2 \left(\ln(x + \sqrt{x^2 + 1}) \right) \cdot \left(\frac{1}{x + \sqrt{x^2 + 1}} \right) \left(1 + \frac{2x}{2\sqrt{x^2 + 1}} \right)$$

$$= \frac{2 \ln(x + \sqrt{x^2 + 1})}{\sqrt{x^2 + 1}}$$

$$y_1 \sqrt{x^2 + 1} = 2 \ln(x + \sqrt{x^2 + 1})$$

$$y_1^2 (x^2 + 1) = 4 \left(\ln(x + \sqrt{x^2 + 1}) \right)^2$$

$$y_1^2 (x^2 + 1) = 4y$$

diff. w.r.t. x

$$y_1^2 (2x) + (x^2 + 1) \cdot 2y_1 y_2 = 4y_1$$

$$xy_1 + (x^2 + 1)y_2 = 2$$

$$xy_2 + y_1 + (x^2 + 1)y_3 + y_2(2x) = 0$$

$$(x^2 + 1) \frac{d^3 y}{dx^3} + (3x) \frac{d^2 y}{dx^2} + \frac{dy}{dx} = 0$$

$$p = 3, q = 0, r = 1, s = 0$$

$$\Rightarrow p + q + r + s = 4$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

01-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS, TNAS & TAAS]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B	A,B	A,B,D	B,C	A,C	A
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	B	A	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	4.00	-46.00	19.67	2.00	4.00
SECTION-II (ii)	Q.	7	8	9			
	A.	6	4	3			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C,D	A,B,C,D	D	A,B	B,C,D	B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	D	A	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	3.00	2.00	0.56	1.25	0.33	2.00
SECTION-II (ii)	Q.	7	8	9			
	A.	3	3	0			

PART-3 : MATHEMATICS

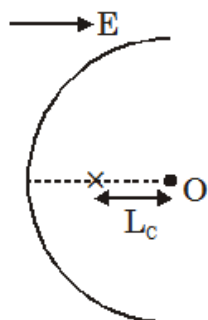
SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,D	A,B,D	A,C,D	B,C,D	A,B,D	B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	D	C	B	A		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	0.00	19.00	1.00	5.00	1.00
SECTION-II (ii)	Q.	7	8	9			
	A.	3	64	2			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B)



$$T = 2\pi \sqrt{\frac{I}{mg \cdot L_c}}$$

$$\text{Put } I = mR^2$$

$$g = \frac{\lambda (\pi R) E}{m}$$

2. Ans (A,B)

A = deformation in equilibrium state.

$$2kA = \frac{\sigma}{\epsilon_0} \cdot q$$

$$\therefore A = \frac{\sigma q}{2k\epsilon_0}$$

Springs are connected in parallel $K_{eq} = 2k$

$$\text{Angular frequency} = \sqrt{\frac{2k}{m}}$$

4. **Ans (B,C)**

By energy conservation $\frac{1}{2}kx^2 = mg(2x)$

$$\Rightarrow x = \frac{4mg}{k} \text{ (maximum elongation)}$$

$$\text{at equilibrium } kx = 2mg \Rightarrow x = \frac{2mg}{k}$$

$$\text{So, (K.E.)}_{\max} = mg(2x) - \frac{1}{2}kx^2$$

$$= 2mg \left(\frac{2mg}{k} \right) - \frac{1}{2}k \left(\frac{2mg}{k} \right)^2$$

$$\frac{1}{2}mv_{\max}^2 = \frac{2m^2g^2}{k}$$

$$\Rightarrow v_{\max} = 2g\sqrt{\frac{m}{k}}$$

5. **Ans (A,C)**

$$\frac{\mu_1}{v} - \frac{\mu_2}{u} = \frac{\mu_1 - \mu_2}{R}$$

$$\Rightarrow \frac{\mu_1}{v} - \frac{\mu_2}{u} < 0$$

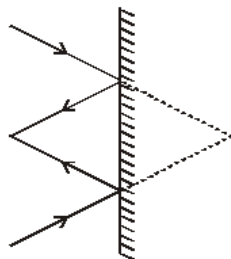
$$\Rightarrow \frac{\mu_1}{v} < \frac{\mu_2}{u}$$

$$\Rightarrow \frac{v}{u} > \frac{\mu_1}{\mu_2}$$

$$\Rightarrow \frac{v}{u} > 0$$

$\Rightarrow v$ and u must have same sign.

6. **Ans (A)**

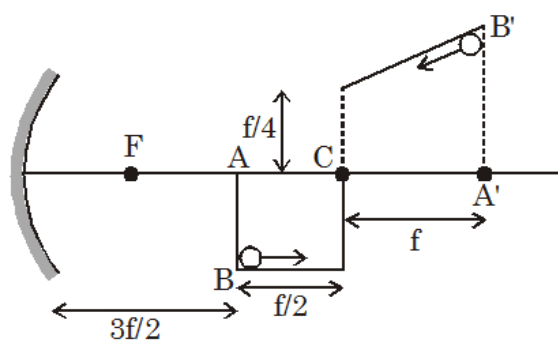


PART-1 : PHYSICS

SECTION-I (ii)

9. **Ans (A)**

The image of the object can be easily drawn by using the mirror formula. Right upright side of wire frame is on centre of curvature so its image would be real, inverted and of same size forming at the centre of curvature. For left upright side i.e. for point A. Use mirror formula.



$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} - \frac{1}{3f} = -\frac{1}{f} \Rightarrow v = -3f$$

i.e. image of A forms at $-3f$ distance from pole.

Height of image of AB can be computed by using

$$\text{magnification formula } m = \frac{h_1}{h_0} = -\frac{v}{u}$$

$$\Rightarrow h_i = - \left[\frac{-3f}{-\frac{3f}{2}} \right] \times \left[-\frac{f}{4} \right] = \frac{f}{2}$$

The top points of upright section of image of wire frame would be joined by a straight line, as the image of linear object would be always straight.

The particle always remain in contact with the lower edge of wire frame so the image of it moves along the slant part of the image.

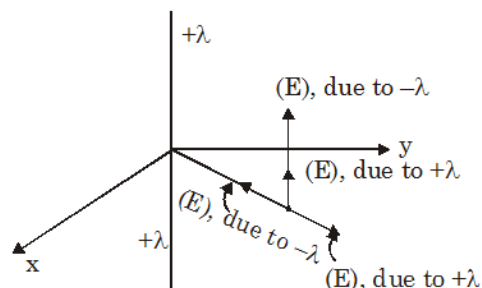
10. **Ans (D)**

It is clear from ray diagram that the velocity of image of particle is varying.

PART-1 : PHYSICS

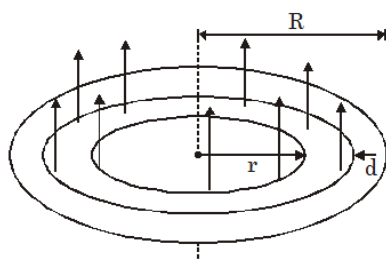
SECTION-II (i)

1. Ans (2.00)



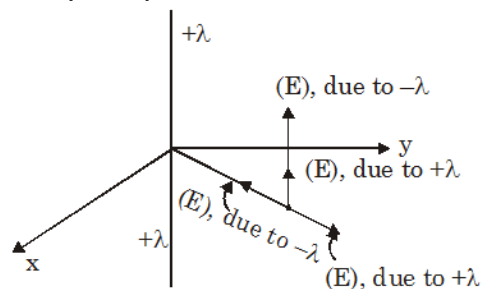
$$E_{\text{net}} = \frac{2k\lambda}{r} = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$\phi_{\text{Net}} = \int_0^R \frac{\lambda}{2\pi\epsilon_0 r} \cdot 2\pi r \cdot dr$$



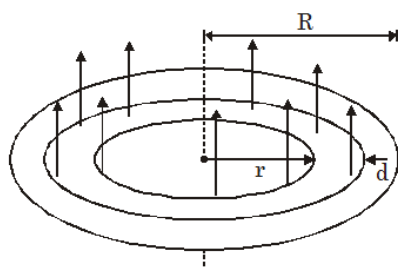
$$\phi_{\text{Net}} = \frac{\lambda}{\epsilon_0} R$$

2. Ans (4.00)



$$E_{\text{net}} = \frac{2k\lambda}{r} = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$\phi_{\text{Net}} = \int_0^R \frac{\lambda}{2\pi\epsilon_0 r} \cdot 2\pi r \cdot dr$$



$$\phi_{\text{Net}} = \frac{\lambda}{\epsilon_0} R$$

3. Ans (-46.00)

$$\vec{p}_i = 1(8\hat{i} + 6\hat{j}) + 2(5\hat{i} - 5\hat{j}) = 18\hat{i} - 4\hat{j}$$

$$\vec{p}_i = \vec{f}_{\text{ext}} \Delta t + \vec{p}_{i1} = -60\hat{j} + 18\hat{i} - 4\hat{j} = 18\hat{i} - 64\hat{j}$$

4. Ans (19.67)

Initial velocity of centre of mass

$$u_{\text{cm}} = \frac{\vec{p}_i}{m_1 + m_2} = \frac{18\hat{i} - 4\hat{j}}{3} = 6\hat{i} - \frac{4}{3}\hat{j}$$

$$\therefore \vec{r}_{\text{cm}(t=1)} = \vec{r}_{\text{cm}(t=0)} + \vec{u}_{\text{cm}}(1) + \frac{1}{2}\vec{a}_{\text{cm}}(1)^2$$

$$= 20\hat{j} + \left(6\hat{i} - \frac{4}{3}\hat{j}\right) + \frac{1}{2}(-10\hat{j}) = 6\hat{i} + \frac{41}{3}\hat{j}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (6)

$$U_i + k_i = U_f + k_f$$

$$-\frac{kq^2}{5^2} + 1.5 = -\frac{kq^2}{r^2} + 0$$

$$r = 6\text{m}$$

8. Ans (4)

$$U = 2\left(-\frac{kQ^2}{r}\right) + \frac{kQ^2}{2r}$$

$$= \frac{kQ^2}{r}\left(-2 + \frac{1}{2}\right) = -\frac{3}{2}\left(\frac{kQ^2}{r}\right)$$

$$\frac{kQ^2}{r^2} - \frac{kQ^2}{4r^2} = \frac{mv^2}{r}$$

$$\frac{3kQ^2}{4r^2} = \frac{mv^2}{r} \Rightarrow mv^2 = \frac{3kQ^2}{4r}$$

$$\text{T.E.} = -\frac{3}{2}\frac{kQ^2}{r} + \frac{3kQ^2}{4r}$$

$$= \frac{kQ^2}{r}\left(-\frac{3}{2} + \frac{3}{4}\right) = \frac{kQ^2}{r}\left(-\frac{3}{4}\right)$$

9. Ans (3)

$$F = 10 \times 12 = 120 \text{ N}$$

$$F = kx = 2400 x$$

$$\therefore x = \frac{1}{20}$$

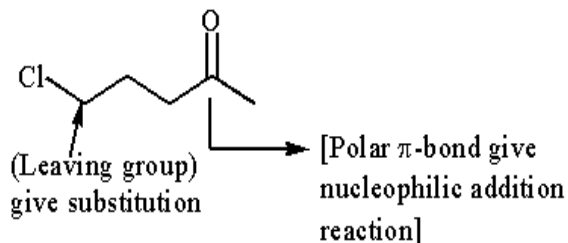
$$\text{Energy stored in the spring } E = \frac{1}{2}kx^2$$

$$= \frac{1}{2} \times 2400 \times \frac{1}{400} = 3\text{J}$$

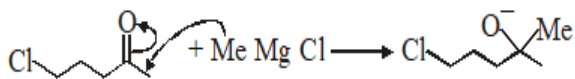
PART-2 : CHEMISTRY

SECTION-I (i)

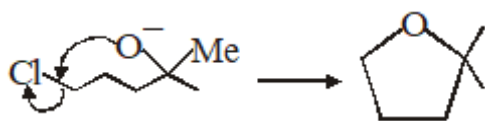
2. Ans (A,B,C,D)



(i) Grignard prefer to give nucleophilic addition on polar π -bond and form anion intermediate.



(ii) In next step anion give intramolecular nucleophilic substitution reaction & form 5 membered ring.



4. Ans (A,B)

$\Rightarrow$ extent of synergic bonding increases as the $-ve$ charge increases on metal

$\Rightarrow$ when synergic bonding is high, M-C bond length decreases, C-O bond length increases.

5. Ans (B,C,D)

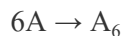
B, C & D options are correct

PART-2 : CHEMISTRY

SECTION-I (ii)

10. Ans (D)

$$n_A = 10 ; n_B = 20 ; n_C = 5$$



$$-\frac{1}{6} \frac{d[A]}{dt} = k = 0.6 ; -\frac{dA}{dt} = 0.6 \times \frac{1}{6} = 0.1$$

$$A_t = A_0 - k \times t$$

$$= 10 - 0.1 \times 60 = 4 \text{ moles of } A_2$$

$$\text{now 4 moles of Aliq.} = 20 ; n_C = 5$$

$$P(s) = \frac{4}{30} \times 100 + \frac{20}{30} \times 90 = \frac{40 + 180}{3} = \frac{220}{3}$$

After 100 min 'A' will finish and convert to A_6

$$\eta_A = \frac{10}{6} = \frac{5}{3} ; \eta_A = 20 ; ; \eta_A = 5$$

$$\eta_T = \frac{5}{3} = 20 + 5 \Rightarrow \frac{5 + 60 + 15}{3} = \frac{80}{3}$$

$$P(s) = \frac{5/3 \times 100 + 20 \times 90}{80/30}$$

$$= \frac{5800}{80} = 73.75$$

PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (3.00)

(ii), (iii) (iv)

2. Ans (2.00)

(i), (v)

4. **Ans (1.25)**

$$-\frac{d[A]}{dt} = \frac{1}{2} \frac{d[B]}{dt} = k[A]$$

$$\text{at } t = \infty; [B] = 2M$$

$$\text{so } A_0 = 1M$$

$$\text{when } [B] = 1.2$$

$$[A] = 0.4 \text{ \& } \frac{d[B]}{dt} = 1$$

$$\text{so } \frac{d[A]}{dt} = \frac{1}{2} = K \times (0.4)$$

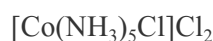
$$\frac{1}{2} = K \times \frac{2}{5}$$

$$K = \frac{5}{4}$$

$$t_{1/2} = \frac{0.7 \times 5}{5} = 0.56$$

$$r = K \times A = \frac{5}{4} = 1.25$$

5. **Ans (0.33)**



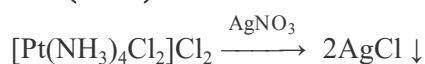
$$x = 3$$

$$y = 6$$

$$z = 3$$

$$\Rightarrow \frac{3}{3+6} = 0.33$$

6. **Ans (2.00)**



PART-2 : CHEMISTRY

SECTION-II (ii)

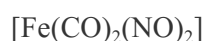
8. **Ans (3)**



9. **Ans (0)**



$$EAN = 26 - 0 + 10 = 36$$



$$EAN \Rightarrow 26 - 0 + 2 \times 2 + 2 \times 3 = 36$$

$$\text{Difference} \Rightarrow 0$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A,B,D)**

$$e^{\sin(x^2+y^2)} = \tan \frac{y^2}{4} + \sin^{-1}x$$

$$\text{at } x = 0$$

$$\Rightarrow e^{\sin y^2} = \tan \frac{y^2}{4} \Rightarrow y = \pm\sqrt{\pi} \pm \sqrt{5\pi} \text{ etc}$$

on differentiation

$$e^{\sin(x^2+y^2)} \left[\cos(x^2+y^2) \left(2x + 2y \frac{dy}{dx} \right) \right] \\ = \frac{2y}{4} \sec^2 \frac{y^2}{4} \frac{dy}{dx} + \frac{1}{\sqrt{1-x^2}}$$

$$\text{put } x = 0$$

$$\Rightarrow e^{\sin y^2} \left[\cos y^2 \left(2y \frac{dy}{dx} \right) \right] = \frac{y}{2} \sec^2 \frac{y^2}{4} \frac{dy}{dx} + 1$$

$$\text{at } x = 0, y = \sqrt{\pi} \Rightarrow \frac{dy}{dx} = \frac{-1}{3\sqrt{\pi}}$$

$$\text{at } x = 0, y = -\sqrt{\pi} \Rightarrow \frac{dy}{dx} = \frac{1}{3\sqrt{\pi}}$$

$$\text{at } x = 0, y = \sqrt{5\pi} \Rightarrow \frac{dy}{dx} = \frac{-1}{3\sqrt{5\pi}}$$

$$\text{at } x = 0, y = -\sqrt{5\pi} \Rightarrow \frac{dy}{dx} = \frac{1}{3\sqrt{5\pi}}$$

2. **Ans (A,B,D)**

$$(\sin^{-1}x + \sin^{-1}w). (\sin^{-1}y + \sin^{-1}z) = \pi^2$$

possible only if

$$\sin^{-1}x = \sin^{-1}y = \sin^{-1}z = \sin^{-1}w$$

$$= \frac{\pi}{2} \text{ or } \left(-\frac{\pi}{2} \right)$$

$$\Rightarrow x = y = z = w = 1 \text{ or } (-1)$$

$$\text{Maximum value} = \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} = 1 + 1 = 2,$$

$$\text{Minimum value} = \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} = -2 \text{ also it can}$$

take value zero if each entry is 1.

3. **Ans (A,C,D)**

Put $y = x$

$$\Rightarrow f(2x) = 2f(x)$$

$$\text{Differentiating } 2f'(2x) = 2f'(x)$$

Put $x = 1$

$$f'(2) = f'(1)$$

Put $x = 0$ & $y = x$

$$\Rightarrow f(x) + f(-x) = 0 \Rightarrow f \text{ is odd}$$

Put $(y = 2x)$

$$\Rightarrow f(3x) + f(-x) = 2f(x)$$

$$\Rightarrow f(3x) = 3f(x) \Rightarrow f'(3x) = f'(x)$$

$x = 1$

$$\Rightarrow f'(3) = f'(1)$$

$$\text{Similarly } f'(4) = f'(1)$$

4. **Ans (B,C,D)**

$$(A) \text{ Let } \sin^{-1}x = \theta \Rightarrow \sin \theta = x$$

$$f(x) = \sin 2\theta = 2\sin \theta \cos \theta = 2x\sqrt{1-x^2}$$

$$f(0) = 0 \Rightarrow \text{at } x = 0, 1, -1 \text{ (many-one)}$$

$$(B) f(x) + f(-x) = 0 \Rightarrow \text{odd function}$$

$$(C) g(x) = 2x, \text{ continuous in given interval}$$

$$(D) f(x) \text{ is continuous function}$$

5. **Ans (A,B,D)**

$$L = \lim_{x \rightarrow \infty} \sin^{-1} \sin \left(1 + \frac{1}{x^2 + 1} \right) = 1$$

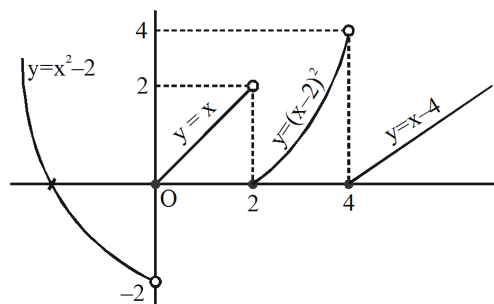
$$(A) \lim_{x \rightarrow \infty} \left[1 + \frac{1}{x^2 + 1} \right] = 1$$

$$(B) \lim_{x \rightarrow \infty} \frac{1 + \frac{2}{x^2}}{1 + \frac{1}{x^2}} = 1$$

$$(C) \lim_{x \rightarrow \infty} \left\{ 1 + \frac{1}{x^2 + 1} \right\} = \lim_{x \rightarrow \infty} \left(\frac{1}{x^2 + 1} \right) = 0$$

$$(D) \lim_{x \rightarrow \infty} \left\{ \frac{x^2}{x^2 + 1} \right\} = \lim_{x \rightarrow \infty} \frac{x^2}{x^2 + 1} = 1$$

6. **Ans (B,C)**



From graph obviously option (B) & (C) are correct.

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (D)**

Pair of straight lines

$$\Rightarrow x^2 - 2xy - y^2 = 0$$

Let point be $(2 + r\cos\theta, 3 + r\sin\theta)$

lies on POSL

$$\Rightarrow (2 + r\cos\theta)^2 - 2(2 + r\cos\theta)(3 + r\sin\theta) - (3 + r\sin\theta)^2 = 0$$

$$\Rightarrow r^2(\cos 2\theta - \sin 2\theta) - r(10\sin\theta + 2\cos\theta) - 17 = 0$$

$$PA + PB = \frac{10\sin\theta + 2\cos\theta}{\cos 2\theta - \sin 2\theta}$$

$$PA \cdot PB = \frac{-17}{\cos 2\theta - \sin 2\theta}$$

$$\frac{d}{d\theta} (PA + PB) = \frac{(\cos 2\theta - \sin 2\theta)(10\cos\theta - 2\sin\theta) - (10\sin\theta + 2\cos\theta)(-2\sin 2\theta - 2\cos 2\theta)}{(\cos 2\theta - \sin 2\theta)^2}$$

$$\text{at } \theta = \frac{\pi}{4}$$

$$\Rightarrow (-1) \left(\frac{8}{\sqrt{2}} \right) - \left(\frac{12}{\sqrt{2}} \right) (-2)$$

$$\Rightarrow \frac{16}{\sqrt{2}} = 8\sqrt{2}$$

8. **Ans (C)**

Pair of straight lines

$$\Rightarrow x^2 - 2xy - y^2 = 0$$

Let point be $(2 + r\cos\theta, 3 + r\sin\theta)$

lies on POSL

$$\Rightarrow (2 + r\cos\theta)^2 - 2(2 + r\cos\theta)(3 + r\sin\theta) - (3 + r\sin\theta)^2 = 0$$

$$\Rightarrow r^2(\cos 2\theta - \sin 2\theta) - r(10\sin\theta + 2\cos\theta) - 17 = 0$$

$$PA + PB = \frac{10\sin\theta + 2\cos\theta}{\cos 2\theta - \sin 2\theta}$$

$$PA \cdot PB = \frac{-17}{\cos 2\theta - \sin 2\theta}$$

$$\frac{-17}{\cos 2\theta - \sin 2\theta} = 17$$

$$\Rightarrow \sin 2\theta - \cos 2\theta = 1$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

9. **Ans (B)**

$$f(x) = \frac{2^x - 1}{2^x + 1} = 1 - \frac{2}{2^x + 1}$$

$$\Rightarrow -1 < f(x) < 1$$

$$\Rightarrow \text{gof is one-one \& has range } \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$$

$$\Rightarrow \text{gof is one-one into function.}$$

10. Ans (A)

$$\lim_{x \rightarrow -\infty} f\left(\sin^{-1}\left(\frac{2^x - 1}{2^x + 1}\right)\right) \Rightarrow \lim_{x \rightarrow -\infty} f\left(-\frac{\pi}{2}\right)^+ = -1$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (2.00)

$$\sin^{-1}x + \sin^{-1}(1-x) = \cos^{-1}x$$

$$\text{domain } 0 \leq x \leq 1$$

$$\sin^{-1}(1-x) = \frac{\pi}{2} - 2\sin^{-1}x$$

$$1-x = \sin\left(\frac{\pi}{2} - 2\sin^{-1}x\right)$$

$$1-x = 1-2x^2$$

$$x = 0, \frac{1}{2}$$

2. Ans (0.00)

$$G(x) = \sin^{-1}(1-x) + \sin^{-1}(1+x) = \pi$$

$$\text{domain has only } x = 0$$

$$G(x) = 0$$

3. Ans (19.00)

$$\frac{dy}{dt} = 2 \text{ cm/sec}, y^3 = -x \Rightarrow 3y^2 \frac{dy}{dt} = -\frac{dx}{dt}$$

$$\Rightarrow \frac{dx}{dt} = -6y^2$$

when $\triangle ABC$ is right triangle

$$\Rightarrow m_{AC} \cdot m_{AB} = -1$$

$$\Rightarrow \frac{y}{x} \cdot \frac{2}{8} = -1 \Rightarrow y = -4x$$

$$y = -4x$$

$$y = \frac{1}{2} \text{ and } x = -\frac{1}{8}$$

(solving with equation $y^3 = -x$)

Area of ABC is

$$\Delta = \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 8 & 2 & 1 \\ x & y & 1 \end{vmatrix}$$

$$= \frac{1}{2} |8y - 2x| = |4y - x|$$

$$= |4y + y^3| = y^3 + 4y \quad (\because y > 0)$$

$$= \frac{d\Delta}{dt} = (3y^2 + 4) \frac{dy}{dt}$$

$$\text{at } t = t_0, y = \frac{1}{2}$$

$$\Rightarrow \frac{d\Delta}{dt} = \left(3 \cdot \frac{1}{4} + 4\right) \cdot 2 = \frac{19}{2} \text{ cm}^2 / \text{sec}$$

4. Ans (1.00)

$$\frac{dy}{dt} = 2 \text{ cm/sec}, y^3 = -x \Rightarrow 3y^2 \frac{dy}{dt} = -\frac{dx}{dt}$$

$$\Rightarrow \frac{dx}{dt} = -6y^2$$

when $\triangle ABC$ is right triangle

triangle

$$\Rightarrow m_{AC} \cdot m_{AB} = -1$$

$$\Rightarrow \frac{y}{x} \cdot \frac{2}{8} = -1 \Rightarrow y = -4x$$

$$y = -4x$$

$$y = \frac{1}{2} \text{ and } x = -\frac{1}{8}$$

(solving with equation $y^3 = -x$)

$$\frac{\beta - \frac{1}{2}}{\alpha + \frac{1}{8}} = \frac{dy}{dx} \bigg|_C$$

$$\frac{(2\beta - 1)4}{8\alpha + 1} = -\frac{1}{3y^2} \bigg|_{y=\frac{1}{2}} = -\frac{4}{3}$$

$$\Rightarrow \frac{2\beta - 1}{8\alpha + 1} = -\frac{1}{3}$$

$$6\beta - 3 = -8\alpha - 1$$

$$6\beta + 8\alpha = 2$$

$$3\beta + 4\alpha = 1$$

5. Ans (5.00)

$g(x)$ will be a quadratic polynomial with roots 1

and -1

$$\text{Let } g(x) = a(x-1)(x+1)$$

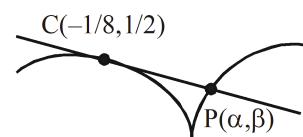
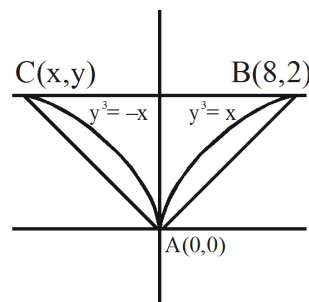
$$\therefore h(x) = \frac{(x-1)(x^3 - 2x^2 - 2x - 2)}{a(x-1)(x+1)}$$

$$\therefore \lim_{x \rightarrow 1} \frac{(x-1)(x^3 - 2x^2 - 2x - 2)}{a(x-1)(x+1)} = \frac{1}{2}$$

$$\Rightarrow -\frac{5}{2a} = \frac{1}{2} \Rightarrow a = -5$$

$$\therefore g(x) = -5(x^2 - 1)$$

Range of $g(x)$: $(-\infty, 5]$



6. Ans (1.00)

$g(x)$ will be a quadratic polynomial with roots 1 and -1

Let $g(x) = a(x-1)(x+1)$

$$\therefore h(x) = \frac{(x-1)(x^3 - 2x^2 - 2x - 2)}{a(x-1)(x+1)}$$

$$\therefore \lim_{x \rightarrow 1} \frac{(x-1)(x^3 - 2x^2 - 2x - 2)}{a(x-1)(x+1)} = \frac{1}{2}$$

$$\Rightarrow -\frac{5}{2a} = \frac{1}{2} \Rightarrow a = -5$$

$$\therefore g(x) = -5(x^2 - 1)$$

$$\lim_{x \rightarrow 0} \frac{5(1 - \cos^2 x)}{5 \ln(1 + \sec^2 x - 1)} = \lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{\sec^2 x - 1} = 1$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (3)

We know that

$$\frac{d^2 y}{dx^2} \cdot \left(\frac{dy}{dx} \right)^{-3} = -\frac{d^2 x}{dy^2}$$

Differentiable w.r.t. y again

$$\frac{d^2 y}{dx^2} (-3) \left(\frac{dy}{dx} \right)^{-4} \cdot \frac{d}{dy} \left(\frac{dy}{dx} \right) + \left(\frac{dy}{dx} \right)^{-3} \frac{d}{dy} \left(\frac{d^2 y}{dx^2} \right) = -\frac{d^3 x}{dy^3}$$

$$\Rightarrow -3 \left(\frac{d^2 y}{dx^2} \right)^2 \left(\frac{dy}{dx} \right)^{-5} + \frac{d^3 y}{dx^3} \left(\frac{dy}{dx} \right)^{-4} + \frac{d^3 x}{dy^3} = 0$$

$$\Rightarrow k = 3$$

8. Ans (64)

$$D = \begin{vmatrix} \alpha - \beta - \gamma & 2\alpha & 2\alpha \\ 2\beta & \beta - \gamma - \alpha & 2\beta \\ 2\gamma & 2\gamma & \gamma - \alpha - \beta \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$D = (\alpha + \beta + \gamma) \begin{vmatrix} 1 & 1 & 1 \\ 2\beta & \beta - \gamma - \alpha & 2\beta \\ 2\gamma & 2\gamma & \gamma - \alpha - \beta \end{vmatrix}$$

$$C_3 \rightarrow C_3 - C_1 \text{ \& } C_2 \rightarrow C_2 - C_1$$

$$D = (\alpha + \beta + \gamma) \begin{vmatrix} 1 & 0 & 0 \\ 2\beta & -(\alpha + \beta + \gamma) & 0 \\ 2\gamma & 0 & -(\alpha + \beta + \gamma) \end{vmatrix}$$

$$D = (\alpha + \beta + \gamma)^3 = 64.$$

9. Ans (2)

$$f(x) = g(x) \cdot \text{sgn}(x^3 + 6x^2 + 11x + 6)$$

$\text{sgn}(x)$ is discontinuous at $x = 0$

$$\left\{ \begin{array}{l} \text{sgn}(x^3 + 6x^2 + 11x + 6) \text{ is discontinuous} \\ \text{when } x^3 + 6x^2 + 11x + 6 = 0 \Rightarrow x = -1, -2, -3 \end{array} \right\}$$

$\Rightarrow f(x)$ to be continuous either $g(-1) = 0$ or $g(-2) = 0$

or $g(-3) = 0$

$$\left. \begin{array}{l} g(-1) = 0 \Rightarrow a = 1 \\ g(-2) = 0 \Rightarrow a = 3/8 \\ g(-3) = 0 \Rightarrow a = 7/27 \end{array} \right\} \begin{array}{l} \text{but a can} \\ \text{take only 1 value} \\ \text{at a time} \end{array}$$

$\Rightarrow$ minimum number of point of discontinuity of $f(x)$

is '2'